

Thermal runaway propagation in linear battery module under low atmospheric pressure

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Abstract

Understanding the mechanism of battery thermal runaway propagation under low atmospheric pressure is critical for the safe operation of battery energy storage systems. This work explores the thermal runaway propagation over a linear arrayed 18650-type lithium-ion battery module in a low-pressure chamber. The effects of ambient pressure (0.1 kPa to 100 kPa), temperature, and electrical connection mode are comprehensively investigated. Results indicate that the propensity of thermal runaway propagation for the open-circuit battery array is much lower, and it only occurs at high ambient temperature and ambient pressure. For parallel-connected battery modules, as ambient pressure decreases, the rate of thermal-runaway propagation first increases due to the reduced environmental cooling (i.e., thermal controlled). It then falls due to lower remaining electrolytes after venting (i.e., venting controlled). The pressure of maximum thermal runaway propagation speed is 60 kPa. The maximum time interval for the thermal runaway of the next cell is about 7 min. A simplified heat transfer analysis was proposed to explain the trend and limits of thermal runaway propagation and reveal the dual effect of pressure. This work provides new insights into thermal runaway propagation, which can deepen the understanding of battery fire safety under low pressure and inspire the thermal-safety design of the lithium-ion battery modules.

Keywords: *Lithium-ion battery; sub-atmospheric pressure; propagation speed; fire hazards*

Nomenclature

Symbols		Subscripts	
A	surface area (m^2)	a	ambient
c	specific heat capacity (J/kg-K)	avg	average
h	heat transfer coefficient ($\text{W/m}^2\text{-K}$)	c	cooling
k	thermal conductivity (W/m-K)	cr	critical condition for thermal runaway
L	length (m)	LIB	lithium-ion battery
m	mass (g)	max	maximum
n	number of cells (-)	min	minimum
P	pressure (kPa)	tab	nickel tabs
$\dot{q}$	heating/cooling rate (kW)	TR	thermal runaway
r	rate (#/min)	v	safety venting
t	time (s)	Abbreviations	
Δt	time interval (min)	CC	constant current
T	temperature (K)	CV	constant voltage
		LIB	lithium-ion battery
Greeks		SOC	state of charge
θ	Temperature difference (K)	SS	stainless-steel

1. Introduction

The global energy sector is experiencing an unprecedented transformation towards a renewable, sustainable, and carbon-neutral future. Featured by high energy density and long cycle life, lithium-ion battery (LIB) is one of the most promising energy storage technologies. However, the fire safety issues of LIB have become increasingly intractable with the continuous improvement of LIB energy capacity, raising public concerns [1–3]. Thermal runaway is the primary concern for battery safety, associated with chemical reactions, gas emissions, smoke, fire, or even explosion [4]. It can be regarded as a rapid self-heating caused by exothermic electrochemical reactions and release tremendous amounts of flammable gases to cause fire incidences [5–7]. The thermal runaway of a single cell can propagate to adjacent cells, escalating the thermal and toxic hazards [8,9]. As thermal runaway propagation is blamed for catastrophic battery fires, understanding its mechanism will greatly contribute to the safe application of large battery systems.

Thermal runaway propagation is fundamentally driven by the heat transfer between cells [10]. In large-format LIB modules constructed by prismatic or pouch cells, heat conduction through the LIB shell is found as the primary heat transfer path for thermal runaway propagation [11–13]. Comparatively,

the dominant heat-transfer mechanism for thermal runaway propagation in cylindrical LIB modules would be more complex because it is more vulnerable to external factors due to the limited contact area between cells [14]. The effects of ambient temperature [15–18], environmental gas composition [18–20], and electrical connection methods [21–23] on thermal runaway and its propagation with cylindrical cells attract extensive research interest. For example, Niu *et al.* [15] found that thermal runaway propagation would speed up at high ambient temperature because of the enhancement of net convective heat transfer between cylindrical cells. Said *et al.* [19] conducted thermal runaway tests for cylindrical cell arrays in nitrogen and air environments. They revealed that the speed of thermal runaway propagation in nitrogen (without flame) was almost nine times slower than that in the air (with flame). Similar results can be found in recent research, where the venting and combustion behaviors of batteries under different oxygen content were comprehensively studied from the perspective of optical assessment [18]. They concluded that the vented oxygen is insufficient for supporting flaming combustion so that the inert environment can reduce the thermal runaway hazards and has the potential to mitigate the thermal runaway propagation. The electrical connections between cylindrical LIBs also affected thermal runaway propagation. For instance, Liu *et al.* [22] showed that the parallel connection accelerated the thermal runaway propagation in the cylindrical LIB module. It is because the heat conduction through tabs is the major path of heat transfer.

With the boosting application of LIBs in low-pressure scenarios, such as plateau, aircraft, and spacecraft, battery safety at low atmospheric pressure has been an emerging topic [24]. The thermal runaway of typical 18650 cylindrical cells has been investigated at low pressure [25–29]. Liu *et al.* [25] observed that a single cell under external heating abuse at low atmospheric pressure behaved similarly to that at standard atmospheric pressure. They also found that low ambient pressure can accelerate the electrolytes escaping from the cell during the venting stage, leading to a weak exothermic reaction and mild thermal runaway. Moreover, the heat release rate of battery fire was found to decrease with ambient pressure owing to the insufficient oxygen supply [26–28]. Previous low-pressure works mainly focused on the fire behavior of a single cell or battery pile. Still, thermal runaway propagation within the LIB module under low pressure is seldom discussed. Recently, Jia *et al.* [30] found that thermal runaway propagation is complex in a low-pressure environment. Reducing the ambient pressure, the thermal runaway propagation rate in the open-circuit LiFePO_4 batteries increases, whereas the thermal runaway propagation rate in the open-circuit $\text{LiNi}_{0.5}\text{Co}_{0.2}\text{Mn}_{0.3}\text{O}_2$ batteries decreases. Although the electric connection of the battery module significantly influences heat transfer-driven thermal runaway propagation, no study has investigated the effects of electric connection and ambient temperature on thermal runaway propagation under low pressure. Therefore, a big knowledge gap imperatively needs to be filled up.

This work aims to (1) explore the thermal runaway propagation over a linear cylindrical LIB module at low pressures (0.1 to 100 kPa), (2) analyze the effects of electric connection, ambient temperature and pressure on thermal runaway propagation; and (3) establish a simplified heat transfer

model to explain the trend and limits of thermal runaway propagation and reveal the dual effect of ambient pressure. This work provides new insights into battery safety and deepens the understanding of LIB fire safety under low pressure.

2. Experimental Setup

3.1. LIB modules

The 18650-type cylindrical cell (ICR18650-22F, Samsung SDI Co., Ltd) is selected in this work, which has a diameter of 18 mm and a height of 65 mm (Fig. 1a). The cell has a nominal capacity of 2.2 Ah, a nominal voltage of 3.6 V, and an initial mass of 41.05 ± 0.05 g. The cathode material of this LIB is lithium nickel manganese cobalt oxide ($\text{LiNi}_{0.5}\text{Co}_{0.2}\text{Mn}_{0.3}\text{O}_2$ or NCM 523). The anode material is made of intercalation graphite. The lithium salt is LiPF_6 , and the solvent of the electrolyte is the mixture of ethylene carbonate (EC), ethyl methyl carbonate (EMC), and dimethyl carbonate (DMC). The separator used in this cell is ceramic coated. According to the specification, the maximum charge current is 2.2 A, and the maximum discharge current is 4.4 A. The operating temperature for the charge is from 0 °C to 45 °C, whereas the temperature for discharge is from -20 °C to 60 °C. The initial internal impedance was measured as 46.2 mΩ by a HIOKI 3561 battery tester.

Prior to the experiments, the state of charge (SOC) for each cell was calibrated to 100% using the battery cycler (Neware BTS, 5V/12A). Specifically, the cells were firstly discharged to 2.75 V (0% SOC) with a constant current of 1.1 A. After the 1-h resting, cells were fully charged (100% SOC) by the Constant Current-Constant Voltage (CC-CV) mode. To avoid the influence of internal heating caused by charging and discharging, all cells were maintained for at least 3 h after the SOC calibration process. LIBs have a very high variability during the production process, which may further affect the experimental results. In this work, the cells from the same batch were employed for charging and discharging to minimize these effects. After SOC calibration and rest process, cells with a similar voltage (4.18 ± 0.02 V) were selected for the experiments. As shown in Fig. 1a, two types of LIB modules were tested in this work. For the open-circuit LIB module, six cells were manually assembled into a linear array where the customized Al_2O_3 ceramic frame was used to hold cells and fix their position with a 1-mm air gap. To acquire the accurate temperature and forestall the external short circuits, the body packaging of each cell was stripped off except for the plastic packaging covering the cathode cap (around 10-mm height). For the parallel-connected LIB module, the nickel strips with the 0.1-mm thickness and 8-mm width were used as tabs (Fig. 1b). A shot welder was employed to connect the nickel strips and cells together, where eight solder joints were produced for one cell to ensure the connection rigidity. Since the welding process would release heat, all LIB modules rested for at least 2 h before experiments to minimize the influence.

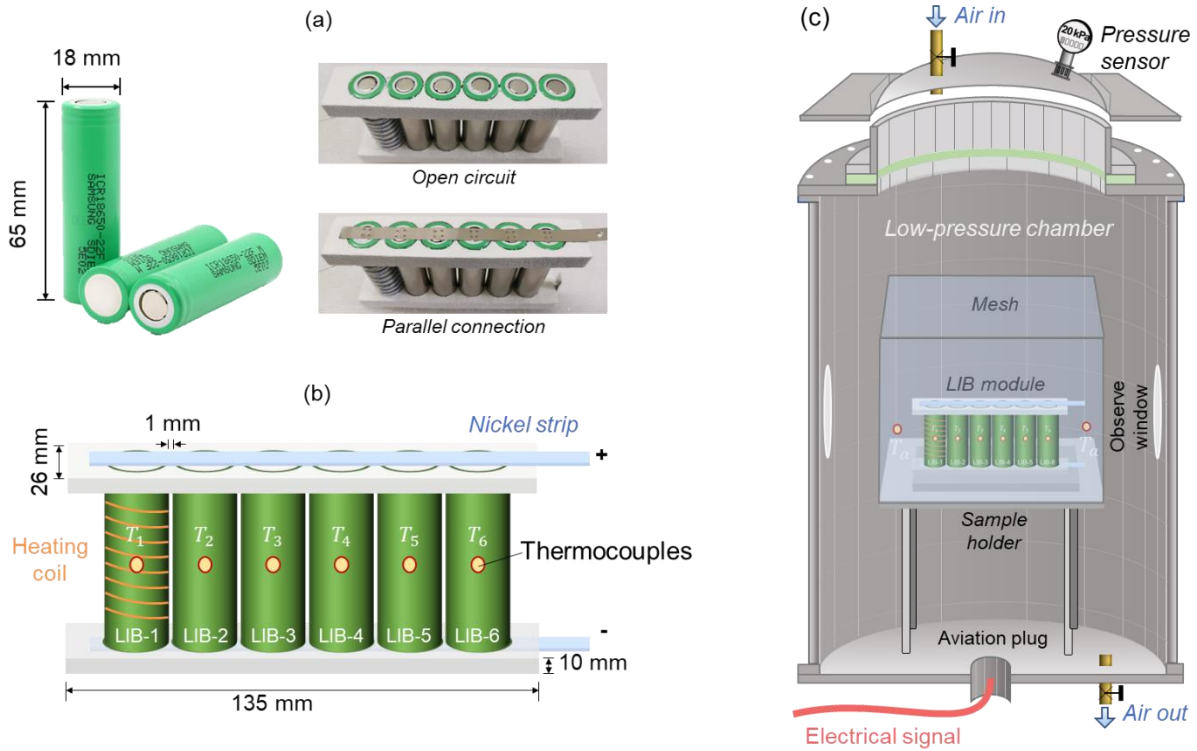


Fig. 1 (a) Photographs of LIB cells and LIB modules tested in this work, as well as the schematic diagrams of (b) parallel-connected LIB module and (c) experimental setup.

3.2. Experimental setup

As illustrated in Fig. 1b, an electric resistance wire ($\text{Cr}_{20}\text{Ni}_{80}$) with a diameter of 0.25 mm was used to trigger the thermal runaway of the first battery (LIB-1). To avert the short circuits, the heating coiling was inserted inside the high-temperature fiberglass sleeving and then tightly and uniformly convolved around the LIB-1. The heating was controlled by a regulated power supply, and the heating power was constant at 30 W for all tests. The tiny K-type thermocouples with the 0.2-mm bead and an accuracy of 0.1 °C were used to measure the cell surface temperature at half height. To acquire the accurate surface temperature of LIBs, the plastic packaging of each cell was stripped off, where the thermocouple bead was fixed on the half-height of the cell surface by a Teflon tape of 10-mm width. This Teflon tape can withstand temperatures higher than 300 °C. To further enhance the contact between the thermocouple and LIB, a thin stainless-steel (SS) wire (0.2-mm diameter) was employed to further fix the Teflon tape and thermocouple bead on the cell surface. The temperature information of cells was collected by a data logger (HIOKI LR8400), with a sampling frequency of 1 Hz throughout the test.

All experiments were conducted in a specially designed pressure chamber with an internal volume of 260 L. This chamber was made of a SS panel and coated with heat-insulating paint, which can provide a well-controlled environment for this work. Fig. 1c shows the cutaway view of the chamber where the internal pressure was achieved by actively controlling the air inlet and air outlet. A vacuum pump with an extraction rate of 600 L/min and an air compressor were responsible for the air extract and supply,

respectively. The minimum pressure inside the chamber was about 0.1 kPa. Inside the pressure chamber, the LIB module was placed on the insulated sample holder where the cuboidal SS mesh was fixed around the LIB to prevent the chamber from fire and explosion. To ensure airtightness, all cables are connected to the aviation plug and passed through the chamber bottom. The internal pressure was monitored by a pressure sensor (Meacon MIK-P300G). The internal ambient temperature (T_a) was measured by two K-type thermocouples (Fig. 1c).

3.3. Test parameters and procedures

Before sealing the low-pressure chamber, the LIB module was carefully placed on the sample holder, and the working state of the measurement devices was checked. Afterward, the vacuum pump was turned on until the ambient pressure decreased to the desired value. Once the targeted pressure was reached, it would be stabilized for at least 10 min. Then, the heating was initiated, and all measurements were started. During the tests, three key parameters are controlled.

- 1) **Tab connection method.** Two types of LIB modules were selected in this work, including the open-circuit module (i.e., without tab connection) and the parallel-connection module (Fig. 1a). The flat nickel tab is widely used to parallelly connect cells for LIB systems with the merits of low cost and easy production. As the parallel LIB module shows a relatively high fire risk at standard pressure, the thermal runaway propagation in the LIB module with parallel-connection mode was mainly discussed in this study.
- 2) **Ambient temperature (T_a).** Previous studies suggested that thermal runaway propagation is sensitive to ambient temperature at standard pressure. Thereby, experiments were carried out at the ambient temperatures of 23 ± 2 °C (spring) and 33 ± 2 °C (summer) under low pressure.
- 3) **Ambient pressure (P).** Systematically investigating the pressure effect on thermal runaway propagation is an important purpose of this work. Six different ambient pressures, 0.1 kPa, 20 kPa, 40 kPa, 60 kPa, 80 kPa, and 100 kPa (close to standard pressure) were adopted to evaluate the characteristics of thermal runaway propagation in the LIB module.

The non-uniformity of cells and the ambient pressure increase caused by gas released during the thermal runaway process are sources of experimental uncertainty. Preliminary tests showed that the thermal runaway of the fully charged cell increased the pressure by 2 kPa and then gradually decreased to the original level. Thus, the chamber is large enough to ignore the temporary and slight pressure rise caused by thermal runaway. LIB cells in this work were selected from the same batch and behaved with similar initial voltage and mass, so that the influence of manufacturing differences was minimized. Considering the limitation of available cell numbers and the toxicity of thermal-runaway tests, we adopted at least two repeating tests for each case and more than three repeating tests near the critical conditions to reduce random error and sustain good accuracy of experimental results.

3. Experimental Results

3.4. Effects of electrical connection

Fig. 2 compares the evolution of LIB surface temperature and post-test photos for (a) parallel-connection module and (b) open-circuit module. The ambient temperature is 33 ± 2 °C, and the ambient pressure is close to the standard atmospheric pressure (100 kPa). It is apparent that the thermal runaway of cells in the LIB module occurred in sequence. In other words, the thermal runaway propagated through the entire LIB module. For each cell, two critical events of safety venting and thermal runaway separated the failure process into three stages, including the (I) pre-heating, (II) venting, and (III) thermal runaway. Taking LIB-1 in Fig. 2a (black line) as an example, its surface temperature escalated with an average rate of 0.4 °C/s during the pre-heating stage. At around 6 min, a slight temperature decrease is observed when the safety vent opens to cause a rapid heat release.

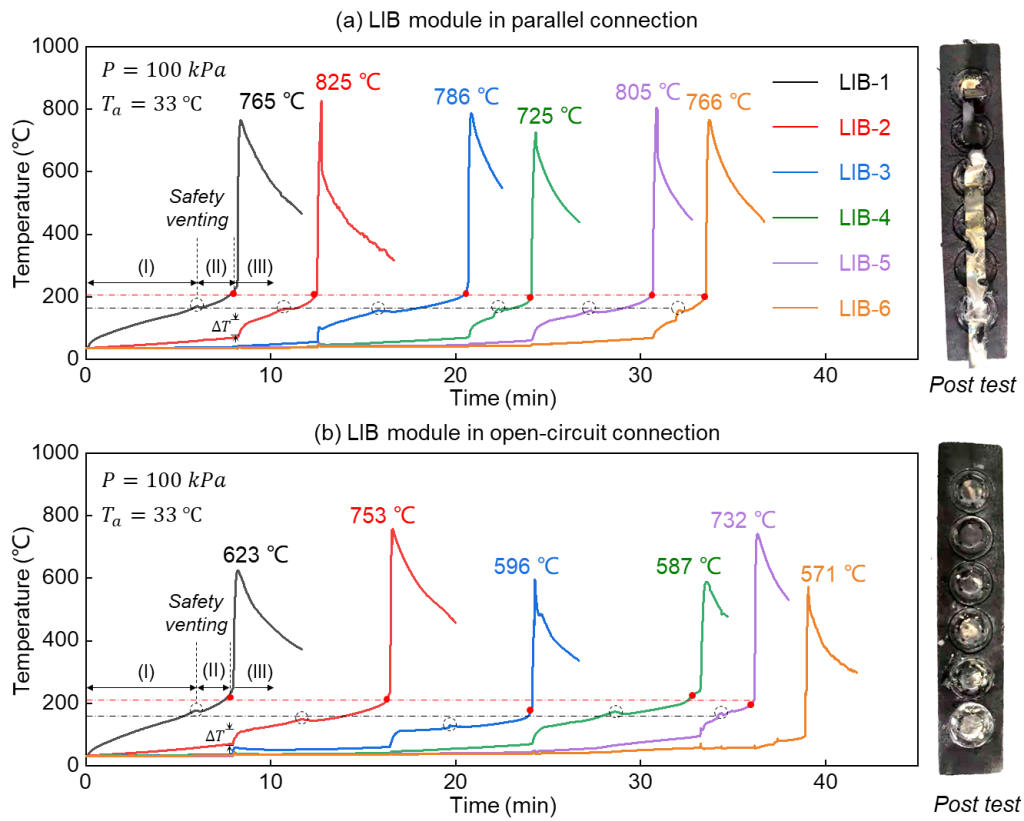


Fig. 2. The temperature evolution for cells in (a) parallel connection module and (b) open-circuit module at the ambient temperature of 33 °C and the ambient pressure of 100 kPa.

During the venting stage, the self-heating reactions inside the cell continued in progress and further increased its temperature. After another 2 min, the thermal runaway was triggered by violent reactions. Then, the LIB temperature rapidly raised from 216 °C to 765 °C, accompanied by large amounts of gases and spark ejection. After the test, all cells burnt out and were covered by the black materials produced by electrochemical and combustion reactions. The aforementioned process is similar to

thermal runaway behaviors of cylindrical cells reported in many studies [31–33]. Particularly, safety vent bursting is a signal of LIB failure where the onset temperature (T_v) is closely related to the cathode material and the SOC level [34,35]. For cells with 100% SOC in Fig. 2, the onset temperature for safety venting (T_v) is around 162 ± 12 °C. This value is 25 °C lower than that of the same cells with 30% SOC [25], so the thermal stability of high-SOC cells is relatively poor. Note that the opening of the safety vent depends on set pressure that has not been clearly defined. Since several studies revealed that the internal pressure could exceed 1.2 MPa for a cylindrical cell during thermal runaway, threshold pressure value ranging from 1.3 MPa to 1.6 MPa has been widely selected by manufacturers [35].

Despite similar thermal behaviors, the whole propagation time in the parallel-connected module (about 25 min) was much shorter than the 30 min in the open-circuit module. Such a result could be attributed to the enhancement of heat transfer by nickel tabs. For instance, when thermal runaway occurred in LIB-1, the temperature of LIB-2 in the parallel-connected module increased from 69 °C to 118 °C with an increment of 49 °C in 1 min (Fig. 2a). However, it only increased 35 °C in 1 min in the open circuit module, indicating a weaker heat transfer (Fig. 2b). After thermal runaway, the cells in the parallel-connected module had a maximum surface temperature of $T_{max} = 779 \pm 35$ °C that is higher and more stable than $T_{max} = 643 \pm 78$ °C for the open-circuit module (although the thermocouple-3 in Fig. 2b might have contact problems and will be excluded in the later discussion). Such a trend is similar to previous works on cylindrical cells [10,15], where the cells in parallel connection exhibit a higher fire hazard. However, it is different from the observation in the large format battery module [36,37], where the peak temperature in connected cells was lower, because the discharging reduced the SOC. In other words, there was no clear discharging process during thermal runaway for connected cylindrical cells, because the tab was damaged [22]. In short, the parallel-connected cylindrical LIB module can lead to a more aggressive thermal runaway propagation than the open-circuit module.

3.5. Effects of ambient temperature

Fig. 3a presents the cell temperature evolution of the parallel-connected module at 23 °C under the ambient pressure of 100 kPa. The thermal runaway did not propagate through the entire module compared to the failed LIB module with the same connection method and ambient pressure in Fig. 2a. Although the LIB-1 and LIB-2 reached thermal runaway with maximum temperatures of 696 °C and 842 °C, the thermal runaway of LIB-3 was not initiated. When the thermal runaway of LIB-2 occurred, the surface temperature of LIB-3 rapidly increased to a local peak of 111 °C from 64 °C. Such a process was similar to the scenario at an ambient temperature of 33 °C in Fig. 2a. However, the temperature increase rate of LIB-3 at the ambient temperature of 23 °C was relatively slower (Fig. 3a). Although the safety venting was also triggered at 16 min, the thermal runaway did not occur in this case. Thus, a lower ambient temperature can reduce the propensity for thermal runaway propagation.

Fig. 3(b-c) further compares the thermal-runaway characteristics of the parallel-connected module at different ambient temperatures (23 °C and 33 °C) at the ambient pressure was 80 kPa. It is interesting

that thermal runaway propagation occurred at 80 kPa. Compared with tests at 100 kPa (Fig. 3a), the environmental cooling could be weaker at 80 kPa due to the lower air density. Thus, it would be easier for cells to accumulate heat and initiate thermal runaway. As this section focuses on the temperature effect, the pressure effect will be discussed later.

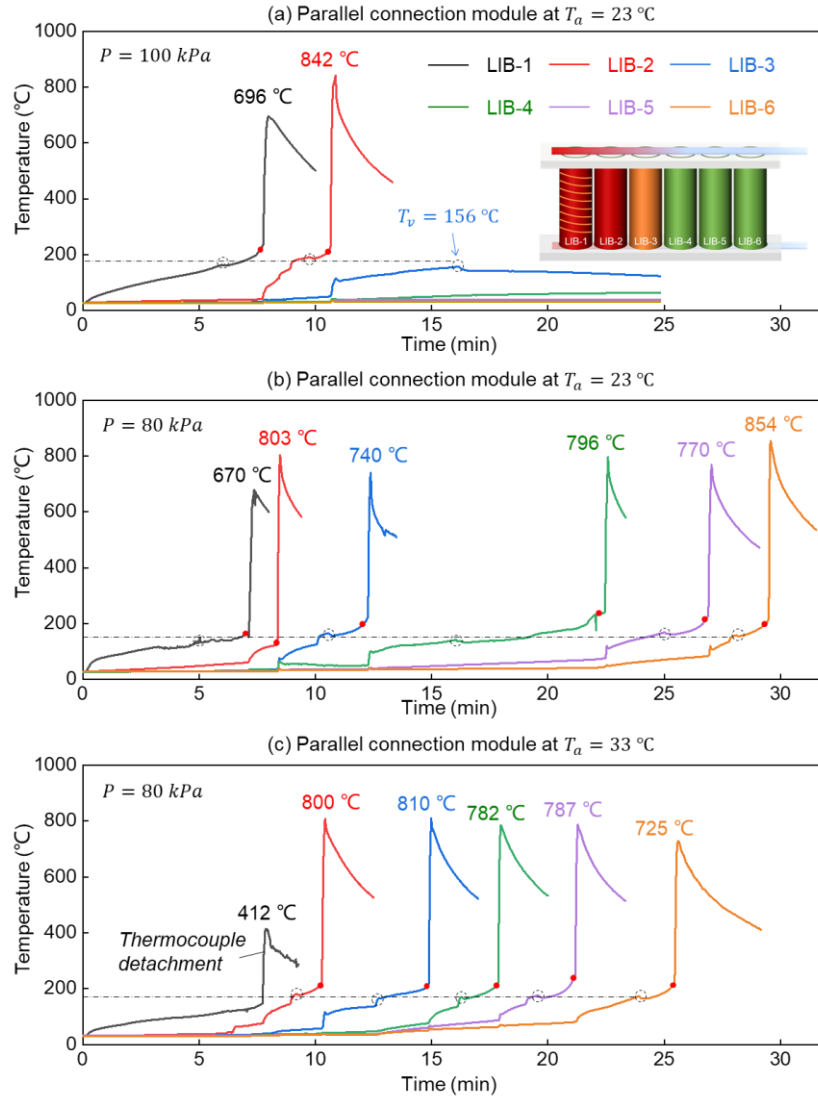


Fig. 3. The temperature evolution for cells in the parallel-connected module at the ambient temperature of (a) 23 °C (100 kPa), (b) 23 °C (80 kPa), and (c) 33 °C (80 kPa).

When the ambient temperature increases, the thermal runaway propagation in the LIB module increases as well, agreeing with the studies at standard pressure [15]. At the ambient temperature of 33 °C (Fig. 3b), the total thermal-runaway propagation time from LIB-1 to LIB-6 was 18 min, so it was 4 min (20%) shorter than that at 23 °C (Fig. 3c). Also, the safety venting temperature seemed insensitive to the ambient temperature. Noted that due to the poor contact of thermocouple, the maximum temperature for LIB-1 in Fig. 3c was underestimated to 412 °C, so it was excluded in the analysis.

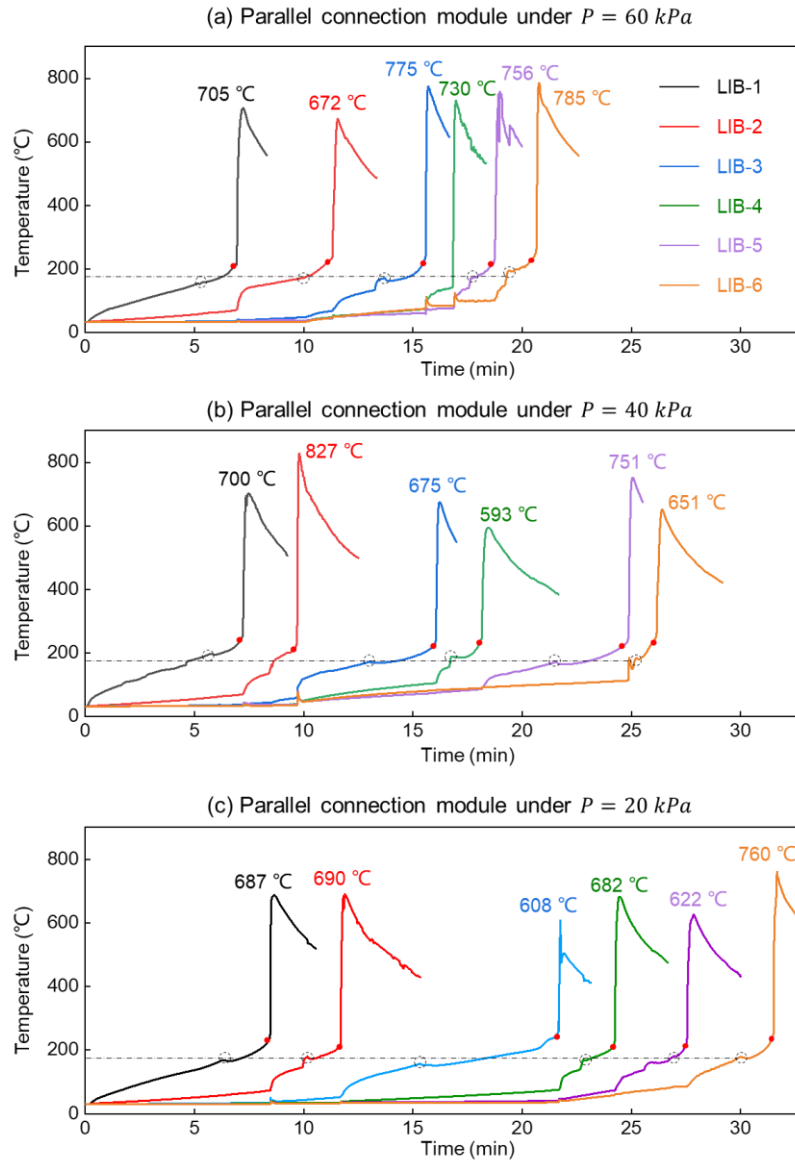


Fig. 4. Temperature curves at the ambient temperature of 33 °C for parallel-connected modules under (a) 60 kPa, (b) 40 kPa, and (c) 20 kPa.

3.6. Effects of ambient pressure

Fig. 4 shows the characteristics of thermal runaway propagation for parallel-connected modules under 20 kPa, 40 kPa, and 60 kPa at the ambient temperature of 33 °C. For the LIB module under 60 kPa (Fig. 4a), the thermal-runaway propagation lasted around 14 min. As the ambient pressure decreased from 60 kPa to 40 kPa, the propagation duration enlarged to 19 min with an increment of 36% (Fig. 4b). Further reducing the ambient pressure to 20 kPa, the whole thermal-runaway propagation time became 23 min. Interestingly, the duration under 60 kPa was also shorter than that under the ambient pressure of 100 kPa (Fig. 2a) and 80 kPa (Fig. 3c). In terms of flame, above 60 kPa, there was a stable flame on the top of the cell after the thermal runaway, while below 60 kPa, no flame was observed on the cell. At 60 kPa, the chance of observing the flame was random. The cathode material of LIB is the NCM523, which can generate oxygen during its decomposition reactions. Thus, the

experimental observations further confirm that the generated oxygen during thermal runaway is insufficient to support the flaming combustion, agreeing with the observed phenomena reported by the previous study [18]. Moreover, T_{max} is lower at a lower ambient pressure, consistent with past works [28].

It is commonly believed that reducing pressure can weaken the flaming combustion and further mitigate thermal runaway propagation. However, no flaming combustion occurred below 60 kPa while the thermal runaway propagation rate still decreases with decreasing ambient pressure. Therefore, there would be another reason brought by low ambient pressure that alleviates the thermal runaway propagation and needs further investigation.

Table 1. The total thermal runaway propagation time for different cases, where “No” represents that thermal runaway did not propagate through the 6 cells of the entire module.

Connection mode	Ambient temperature (°C)	Thermal runaway propagation duration from LIB-1 to LIB-6 (min)					
		100 kPa	80 kPa	60 kPa	40 kPa	20 kPa	0.1 kPa
Open circuit	23	No	No	No	No	No	No
	33	30.5 ± 3.7	No	No	No	No	No
Parallel-connected	23	No	22.2 ± 2.4	16.3 ± 0.7	25.6 ± 3.3	No	No
	33	25.1 ± 1.3	17.1 ± 0.8	13.1 ± 0.6	19.3 ± 1.7	25.7 ± 3.8	No

Table 1 summarizes the total thermal-runaway propagation time in all experimental conditions. In this table, “No” represents that thermal runaway did not propagate through the entire module. For the open-circuit LIB module, the propensity of thermal runaway propagation increases with both the ambient temperature and pressure. For example, thermal runaway propagation did not occur in the LIB module at the ambient temperature of 23 °C while it happened at 33 °C. Reducing the ambient pressure at 33 °C, thermal runaway did not propagate. Thus, reducing the ambient pressure could be an effective choice to mitigate the thermal runaway propagation for NCM batteries in the open-circuit state, agreeing with the research findings delivered by Jia et al. [30].

In terms of parallel-connected LIB modules, high ambient temperature can lead to a much severer thermal-runaway propagation. However, the pressure effect on thermal runaway propagation is more complicated. As the ambient pressure decreases, the propagation time in the parallel-connected module first decreases and then increases. The shortest propagation time occurred at around 60 kPa, which was 16.3 ± 0.7 min at 23 °C and 13.1 ± 0.6 min at 33 °C, respectively. Therefore, the influence of atmospheric pressure on thermal runaway propagation in the parallel-connected LIB module needs further investigation.

4. Discussions

4.1 Temperatures for venting and thermal runaway

The safety venting temperature (T_v), the critical temperature for thermal runaway (T_{cr}), and maximum surface temperature (T_{max}) are extensively used to reflect the thermal stability and thermal hazards of the cell during thermal runaway [38]. T_v was determined as the earliest point where the self-heat rate (dT/dt) became negative. Thermal runaway temperature (T_{cr}) was identified as the temperature point where the value of dT/dt exceeded the 10 °C/s. This criterion considered the side-heating induced thermal runaway propagation and was established by combing visual and audible observations, which was adopted in many studies [39]. And T_{max} is the maximum temperature of thermal runaway, which strongly depends on its cathode chemistry, capacity, electric connection, and environmental cooling condition [38,40]. To highlight the pressure effect, Fig. 5 summarizes the characteristic temperatures of T_v , T_{cr} , and T_{max} for all thermal runaway propagation cases with parallel connections. Note that several thermocouples had contact problems during the tests, so the corresponding data are excluded from the analysis. As shown in Fig. 5a, safety venting temperature (T_v) is insensitive to the ambient temperature and ambient pressure, which is around 166 ± 10 °C. In addition, the onset temperature for thermal runaway (T_{cr}) slightly increases as the ambient pressure decreases. Specifically, when the ambient temperature is 33 °C, T_{cr} increases from 202 °C to 234 °C, as the ambient pressure (P) drops from 100 kPa to 20 kPa. This is because the evaporation and volatilization of electrolytes would be enhanced in the venting stage (i.e., the interval between safety venting and thermal runaway) [41].

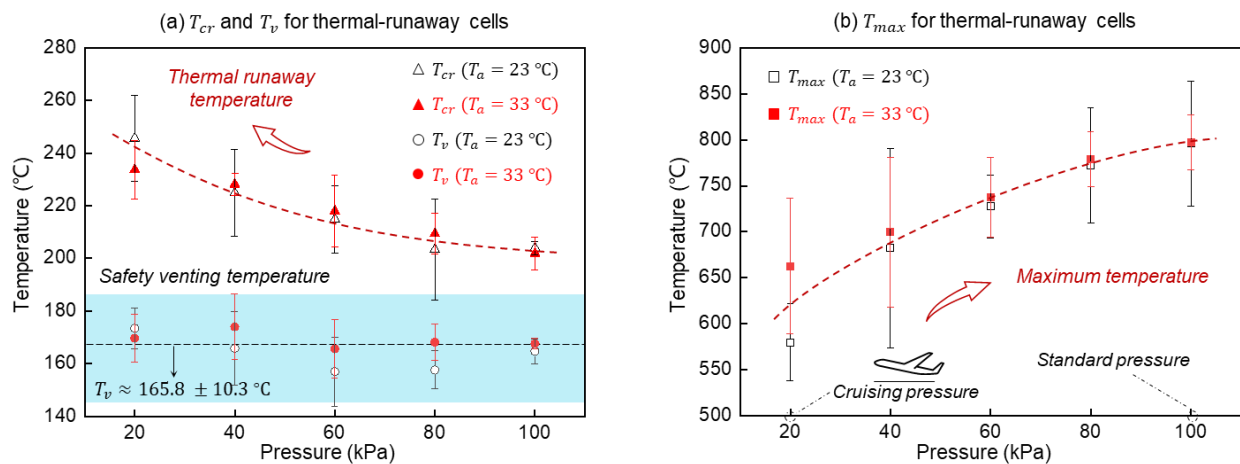


Fig. 5. The characteristic cell temperatures vs. the ambient pressure: (a) safety venting temperature (T_v) and thermal runaway onset temperature (T_{cr}), and (b) maximum surface temperature (T_{max}).

The stronger venting of active electrolytes at lower pressure leads to a lower amount of electrolytes remaining in the cell. Thus, the electrochemical reaction of the remaining electrolyte becomes weaker, and the battery needs a higher temperature to trigger the thermal runaway. Consequently, the maximum

battery temperature after thermal runaway (T_{max}) is lower at a lower ambient pressure [25], as shown in Fig. 5b. Specifically, T_{max} decreases from about 800 °C at 100 kPa (close to standard pressure) to about 600 °C at 20 kPa (the cruising pressure for freighter). The relatively large variance of repeating tests could be primarily ascribed to the differences in the battery itself.

4.2 Thermal-runaway propagation rate

Similar to the fire spread process [42], the thermal-runaway propagation can be viewed as a continuous ignition process of cells in the LIB module. The propagation rate (or speed) can quantify the thermal-runaway hazards in a LIB module. As the thermal runaway propagated from one cell to another successively, a cell-based propagation rate is defined as the number of cells that undergo thermal runaway per minute [15,20] as

$$r = \frac{n}{\Delta t} \quad (1)$$

where Δt [min] is the total time of thermal-runaway propagation, and n [#] is the number of cells that undergo thermal runaway.

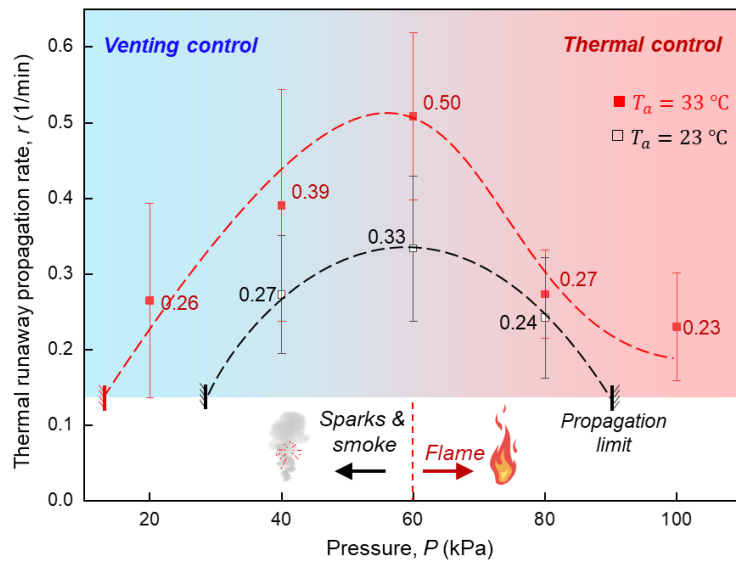


Fig. 6. Thermal-runaway propagation rate in parallel-connected LIB modules under different pressures.

The propagation from LIB-1 to LIB-2 is affected by the heating coil, so it is excluded from the calculation. Then, the thermal-runaway propagation from LIB-2 to LIB-6 was considered to characterize the propagation features. Based on experimental data, the calculated thermal runaway propagation rates from LIB-2 to LIB-6 in parallel-connected modules varying with ambient pressure are summarized in Fig. 6. First of all, the thermal-runaway propagation at 33 °C is 30% faster than that at 23 °C, showing a higher fire risk. At 23 °C, thermal runaway cannot propagate under the ambient pressures of 100 kPa and 20 kPa. At 33 °C, the thermal runaway can still propagate at 20 kPa, but not at 0.1 kPa. Moreover, there is a minimum rate of thermal-runaway propagation of about 0.15 #/min or

a maximum time interval of 7 min. In other words, if the next cell cannot reach thermal runaway in 7 min, the propagation stops. Niu et al. [15] have studied the effects of ambient temperature and SOC on thermal runaway propagation. The LIB module in their work is similar to this work, where six cells were linearly arranged. For parallel-connected LIB modules, thermal runaway can propagate at the ambient temperature of 20 °C. This is because two insulation boards were fixed on both sides of the LIB module in their work, which restricted the convective cooling of cells. The calculated thermal runaway propagation rate for this case is 0.36 #/min, which has a similar magnitude to the propagation rate (0.23 #/min) at 100 kPa in this work. Therefore, the accuracy and precision of the data in Fig. 6 are acceptable.

As the ambient pressure decreases, the thermal runaway propagation rate first increases to a maximum value and then slightly decreases. The maximum propagation rates occur around the ambient pressure of 60 kPa, which are 0.50 #/min at 33 °C and 0.33 #/min at 23 °C, respectively. Therefore, the hazards of thermal runaway propagation within a parallel-connected LIB module under low pressure should not be underestimated. Such a non-monotonical trend suggests that the ambient pressure changes different influence factors that jointly control the thermal-runaway propagation.

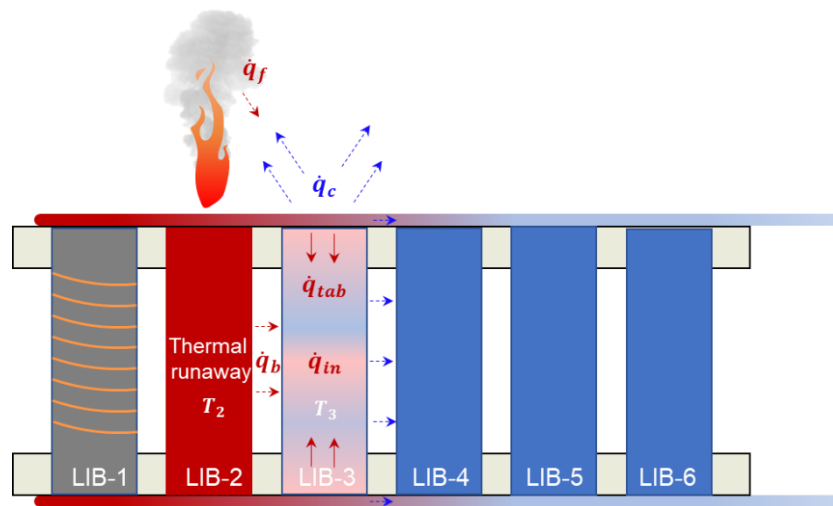


Fig. 7. The heat-transfer illustration for the thermal runaway propagation in a linear LIB module.

4.3 Cooling of thermal-runaway cell

Fig. 7 illustrates the simplified heat transfer processes of thermal runaway from LIB-2 (the thermal-runaway cell) to LIB-3 (the next cell). As discussed above, the second cell (LIB-2) reaches thermal runaway in all cases. If the thermal runaway of LIB-3 (next cell) cannot be initiated by LIB-2 (thermal-runaway cell), the thermal runaway propagation is impossible to occur in the entire module. The cooling process of LIB-2 after thermal runaway is significant for the thermal runaway propagation. Moreover, the air density will decrease as the ambient pressure decreases, which would lead to a lower convective coefficient under low pressure. Thus, the relationship between ambient pressure and the effective cooling coefficient of the environment is vital to explaining the trend of thermal runaway propagation.

Assuming that LIB-2 is primarily cooled by the environment after the thermal runaway, the energy conservation equation is given by

$$m_{LIB}c_{LIB}\frac{dT_2}{dt} = -h_cA(T_2 - T_a) \quad (2)$$

where m_{LIB} is the mass of LIB; c_{LIB} is the total heat capacity of the LIB; T_2 is the temperature of LIB-2; t is the time after the thermal runaway of LIB-2; h_c is the overall cooling coefficient; and A is the cell surface area, respectively. Define the temperature difference $\theta = T_2 - T_a$, then $dT_2/dt = d\theta/dt$. Thus, Eq. (2) can be rewritten as

$$m_{LIB}c_{LIB}\frac{d\theta}{dt} = -h_cA\theta \quad (3)$$

By separating the variables and integrating the Eq. (3) from the initial condition $(0) = T_{max} - T_a$, the correlation between the effective cooling coefficient (h_c) can be expressed as

$$h_c = \frac{m_{LIB}c_{LIB}}{At} \ln \left(\frac{T_{max} - T_a}{T_2 - T_a} \right) \quad (4)$$

According to the previous study conducted by Mao *et al.* [43], the heat capacity of the LIB (c_{LIB}) is 1100 J/kg · K. The mass of LIB (m_{LIB}) used in this work is 0.041 kg, and the surface area of this LIB is $3.67 \times 10^{-3} \text{ m}^2$. Then, the effective cooling coefficient (h_c) can be calculated where the experimental data of the event when thermal runaway of LIB-3 occurred were employed.

Fig. 8a summarizes the calculated value of h_c varying with the ambient pressure (P). As expected, the effective cooling coefficient (h_c) increases from 15 W/m² · K to 27 W/m² · K, as the ambient pressure increases from 20 kPa to 100 kPa. Moreover, the correlation of $h_c \propto P^{0.2}$ agrees with the previous study [25].

4.4 Heat transfer analysis of thermal-runaway propagation

Considering the thermal runaway propagation from LIB-2 to LIB-3 in Fig. 7, the heat sources for LIB-3 include flame heating ($\dot{q}_f$, convection and radiation), the body heating from LIB-2 ($\dot{q}_b$, convection and radiation), the heating from the nickel tabs for parallel-connected cases ($\dot{q}_{tab}$, conduction), as well as the internal heating from the exothermic reactions ($\dot{q}_{in}$). The cooling source of LIB-3 is mainly the environment ($\dot{q}_c$). During the thermal runaway, the jetting flame has upward or lateral directions due to the structure of the venting cap [44]. As the insulation board around the positive cap of LIBs can partially shield the flame that only occurred over 60 kPa, the direct flame heating ($\dot{q}_f$) could be ignored. Before the thermal runaway, the self-heating rate ($\dot{q}_{in}$) of LIB-3 is also negligible. The necessary condition for the continuous propagation is that the battery cell should be heated to its onset temperature for thermal runaway (i.e., T_{cr}).

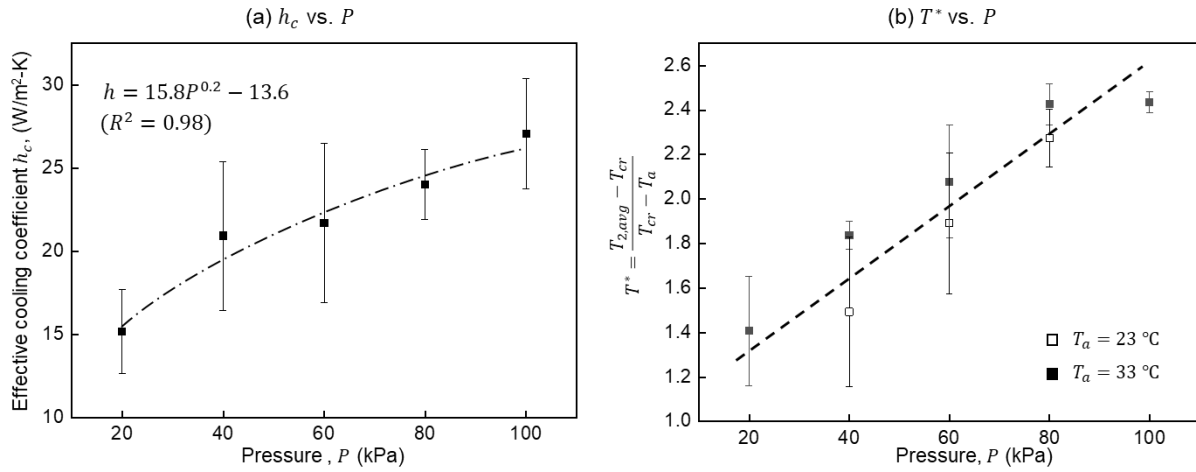


Fig. 8. The effect of ambient pressure (P) on (a) the effective cooling coefficient (h_c) and (b) dimensionless temperature (T^*).

For the open-circuit LIB module, the effective heat transfer between failed cell (LIB-2) and the next cell (LIB-3) is $q_b = h_{TR}A(T_2 - T_{cr})$. The energy conservation equation for thermal runaway of LIB-3 ($T_3 = T_{cr}$) should satisfy

$$h_{TR} \left(\frac{A}{2} \right) (T_2 - T_{cr}) \geq h_c A (T_{cr} - T_a) \quad (\text{open circuit}) \quad (5)$$

where h_{TR} is the overall coefficient of convective and radiant heating from the LIB-2, so the heating area is roughly half of cell surface area. There is no direct contact between cells, so both h_{TR} and h_c are dominated by convection, and they are comparable ($h_{TR} \approx h_c$). The minimum temperature of LIB-2 to trigger the thermal runaway of LIB-3 can be expressed as

$$T_{2,min} = \frac{3T_{cr} - T_a}{2} \quad (\text{open circuit}) \quad (6)$$

According to Eq. (4), the maximum time interval (t_{max}) for the thermal runaway propagation to the next cell should be

$$t_{max} = \frac{m_{LIB} c_{LIB}}{h_c A} \ln \left[\frac{2(T_{max} - T_a)}{3(T_{cr} - T_a)} \right] \quad (\text{open circuit}) \quad (7)$$

which delivers a maximum time interval between cells for thermal runaway propagation. In other words, if the next cell cannot reach thermal runaway within t_{max} , the propagation stops in the open circuit LIB module.

In terms of the parallel-connected module, the heat transfer through the tabs, $\dot{q}_{tab} = kA_{tab}(T_2 - T_{cr})/L$, plays a dominant role in thermal runaway propagation [22]. The necessary condition for thermal runaway propagation should be

$$\frac{kA_{tab}}{L} (T_2 - T_{cr}) \geq h_c A_{LIB} (T_{cr} - T_a) \quad (\text{tab connection}) \quad (8)$$

where k is the thermal conductivity of nickel tabs, A_{tab} is the contact area between tabs and cell, and L is the length of the tab. Then, the minimum temperature of LIB-2 in the parallel-connected module can be expressed as

$$T_{2,min} = \frac{h_c A_{LIB} L}{k A_{tab}} (T_{cr} - T_a) + T_{cr} \quad (\text{tab connection}) \quad (9)$$

Thus, the maximum time interval for thermal runaway propagation can be derived as

$$t_{max} = \frac{m_{LIB} c_{LIB}}{h_c A} \ln \left[\frac{T_{max} - T_a}{\left(\frac{h_c A L}{k A_{tab}} + 1 \right) (T_{cr} - T_a)} \right] \quad (\text{tab connection}) \quad (10)$$

To further explore the dual effect of pressure on thermal runaway propagation in the parallel-connected module, the overall propagation speed (r) is analyzed based on the classical theory of fire spread [42]. Controlling by the ratio of net heating absorption to the thermal inertia of the cell, the overall propagation speed (r) is given by

$$r = \frac{\text{Net heating rate}}{\text{Internal thermal resistance}} \approx \frac{\dot{q}_{tab} - \dot{q}_c}{m_{LIB} c_{LIB} (T_{cr} - T_a)} \quad (11)$$

Using the average temperature of LIB-2 ($T_{2,avg}$) during the propagation stage, it can be rewritten as

$$r \approx \frac{\frac{k A_{tab}}{L} T^* - h_c A}{m_{LIB} c_{LIB}} \quad (12)$$

$$T^* = \frac{T_{2,avg} - T_{cr}}{T_{cr} - T_a} \quad (13)$$

where the dimensionless temperature (T^*) reflects heating intensity. Fig. 8b shows the value of T^* varying with the ambient pressure for the parallel-connected cases. It is evident that T^* increases with the pressure and temperature.

Thus, Eq. (12) elaborates on the effects of ambient temperature and pressure on thermal runaway propagation in the parallel-connected module. Specifically, the thermal-runaway propagation rate increases as ambient temperature (T_a), because of the increasing numerator in Eq. (12). Moreover, reducing the ambient pressure can weaken the environmental cooling and exothermic reactions during the venting stage, resulting in the decrease of both h_c and T^* (Fig. 8).

There is a competitive relationship between T^* and h_c to affect the thermal runaway propagation rate. Specifically, h_c is responsible for the heat-transfer regime (or thermal controlled), whereas T^* is responsible for the chemical regime (or venting controlled). Lowering the ambient pressure from 100 kPa to 60 kPa, the thermal runaway propagation rate slightly increases. Therefore, the reduction of environmental cooling controls the thermal runaway propagation in this regime, while the pressure effect on exothermic reactions is negligible (i.e., thermal controlled in Fig. 6). Further

reducing the ambient pressure, the pressure effect on weakening exothermic reactions can no longer be neglected. When the ambient pressure varies from 60 kPa to 20 kPa, the pressure effect on internal reactions seems to exceed the decrease of heat dissipation. Then, the thermal runaway propagation is controlled by the mitigation of internal reactions and slightly decreases (venting controlled).

It should be noted that the pressure value for maximum thermal runaway propagation rate is only suitable for the 18650-type LIB of a particular material system. According to Eq. (12), the propagation rate strongly depends on the thermophysical characteristics of nickel tabs and LIB cells. Nevertheless, the thermal safety of parallel-connected LIB modules under low pressure should not be ignored. Further research is imperatively needed to explore the pressure effect in LIB modules with different materials, which can help to make the lithium-ion batteries safer in the application.

5. Conclusions

In this work, thermal runaway propagation over a linear arranged cylindrical lithium-ion battery module was tested using a specially designed low-pressure chamber. The effects of connection methods (open-circuit and flat-tab parallel connection), ambient temperature (23 °C and 33 °C), and ambient pressure (0.1 kPa to 100 kPa), were comprehensively investigated. Results indicated that the safety venting temperature (165.8 ± 10.3 °C) for fully charged cells was insensitive to ambient temperature and pressure. The thermal runaway temperature (201 ~ 245 °C) decreased with ambient pressure, while the maximum LIB surface temperature (579 °C ~ 797 °C) increased. The thermal runaway for the open-circuit battery array only occurs at high ambient temperature and pressure.

Compared to the open-circuit module, thermal runaway propagation in the parallel-connected battery module was more aggressive. As atmospheric pressure decreases, the rate of thermal-runaway propagation first increases due to the reduced environmental cooling (i.e., thermal controlled) and then decreases due to lower remaining electrolytes after venting (i.e., venting controlled). The maximum thermal runaway propagation speed is 0.50 #/min, which appears at 33 °C under 60 kPa. The maximum time interval for the thermal runaway of the next cell is about 7 min. Moreover, a simplified heat transfer analysis was proposed to qualitatively elaborate the trend of thermal runaway propagation and reveal the underlying mechanism of the pressure effect. This work reports the thermal-runaway propagation characteristics under low-pressure environments, which could strengthen our understanding of thermal-runaway propagation and provide scientific guidelines for the fire-safe design of lithium-ion battery systems.

CRedit authorship contribution statement

Yanhui Liu: Data curation, Investigation, Writing - Original Draft, Formal analysis. **Huichang Niu:** Conceptualization, Methodology, Resources, Supervision. **Cangsu Xu:** Methodology, Resources. **Xinyan Huang:** Conceptualization, Methodology, Funding acquisition, Supervision, Writing - Review & Editing.

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Appendix

Figure A1 is the photo of the low-pressure chamber. Figure A2 is the temperature evolution curves of parallel-connected LIB module at 0.1 kPa ($T_a = 33\text{ }^{\circ}\text{C}$), where thermal runaway did not propagate through the entire LIB module.

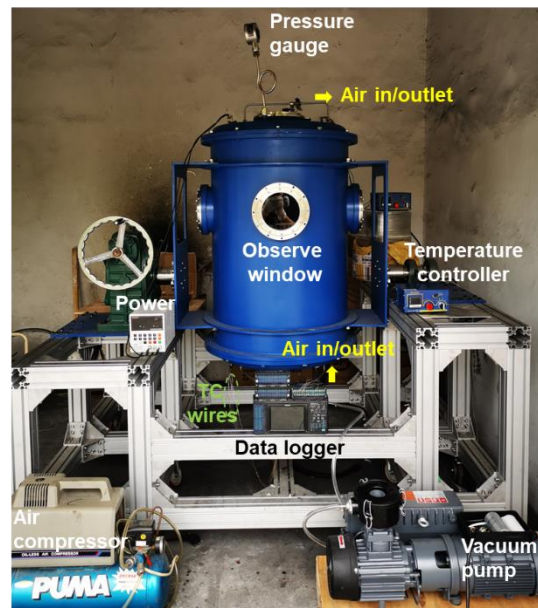


Fig. A1. The photograph of the low-pressure chamber.

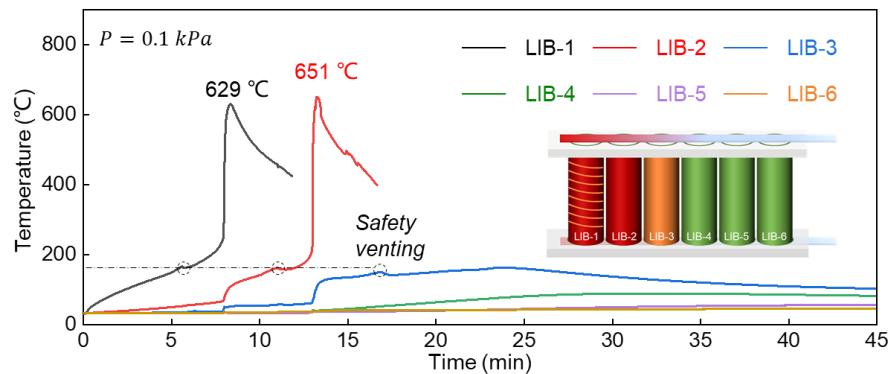


Fig. A2. The temperature evolution for cells in the parallel-connected LIB module at $33\text{ }^{\circ}\text{C}$, where the ambient pressure is 0.1 kPa.